25900399 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCFTQ201 - FOOD ANALYTICAL INSTRUMENTATION

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **5** questions

Each question carries **2** marks

1. What are the applications of chromatography in food industry? [U] / [CO1]
2. Define the mobile phase in GC and give examples. [U] / [CO2]
3. What are the benefits and limitations of starch gel electrophoresis? [K] / [CO5]
4. Define electrophoresis. Who pioneered the technique of electrophoresis? [K] / [CO5]
5. What is the role of centrifugation in dairy industry? [U] / [CO5]
6. What is nuclear magnetic resonance? [U] / [CO4]
7. What is the principle of ICP-AES? [U] / [CO4]

[2x5 = 10]

Part B

Short Essay Questions

Answer any **5** questions

Each question carries **4** marks

8. Add a note on microchip electrophoresis. [K] / [CO5]

9. What is the difference between horizontal and vertical electrophoresis? [U] / [CO5]

10. List out the factors affecting fluorescence. [K] / [CO4]

11. What is a Nernst Glower? [U] / [CO4]

12. In a TLC experiment, two components A and B moved 7 cm and 9 cm respectively from the origin. The solvent front moved 10 cm. Calculate the Rf values for components A and B. [A] / [CO2]

13. Describe the phenomenon of interaction radiation with matter. [U] / [CO3]

14. Give an overview of different detectors employed in HPLC [U] / [CO2]

[4x5 = 20]

Part C

Long Essay Questions

Answer any **2** questions

Each question carries **10** marks

15. Explain the principle, working, and applications of paper chromatography. [U] / [CO2]

16. Explain techniques and working of atomic absorption spectroscopy. [U] / [CO4]

17. Write briefly about differential, rate zonal, and isopycnic centrifugation methods. [U] / [CO5]

18. Describe the different modes of centrifugation. [U] / [CO5]

[10x2 = 20]